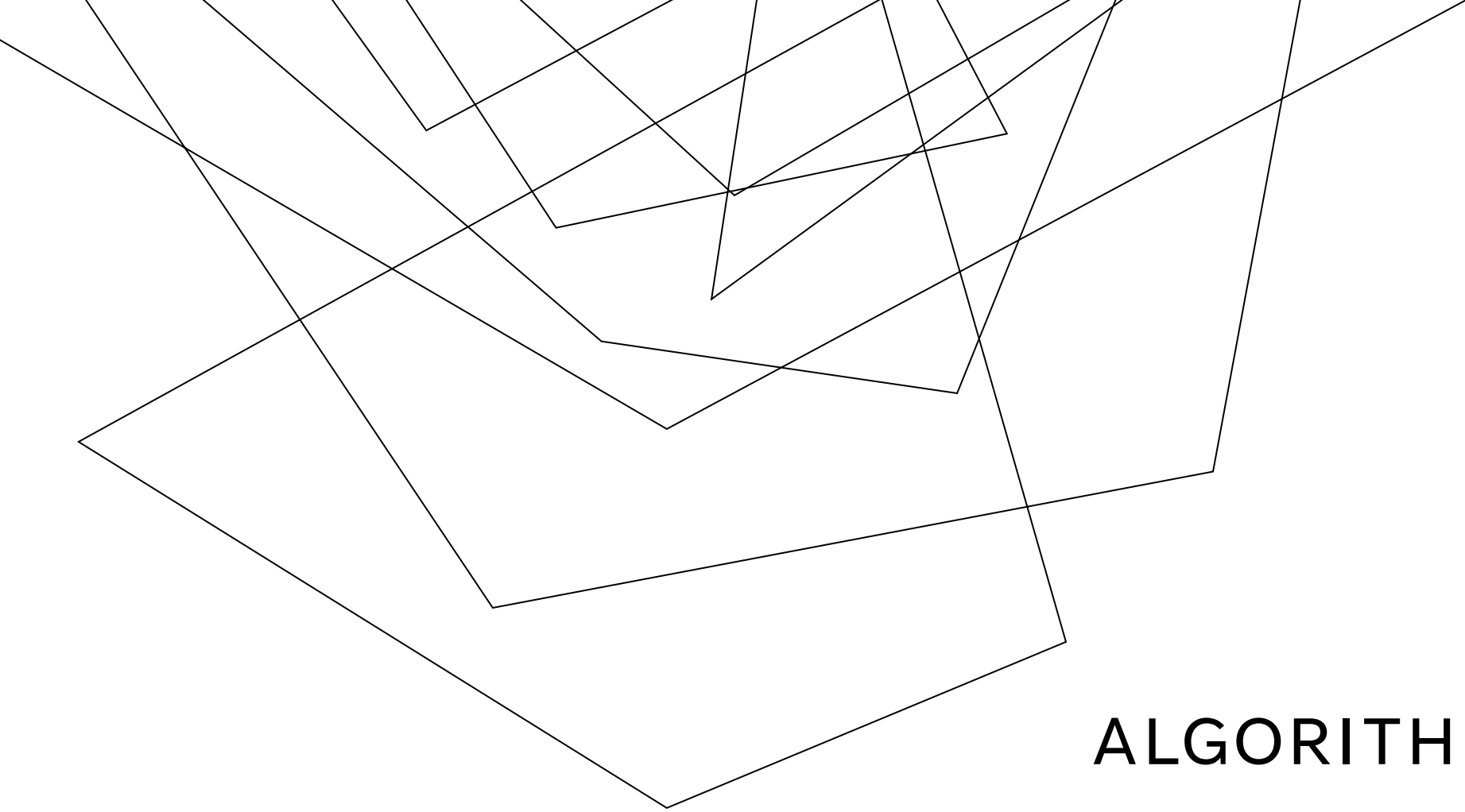


A series of thin, black, overlapping lines forming various geometric shapes like triangles and polygons, creating a complex, abstract pattern in the upper left portion of the slide.

ALGORITHMS

4. SORTING (1)





공지 사항

2026 대학생 프로그래밍 경진대회

접수 관련 학부 홈페이지 참조 (수요일까지 접수)



참가 대상

대한민국 소재 대학교(원)
재학생 3명(3인 1팀) 및 코치교수 1명

※ 자세한 내용은 대외 홈페이지 참가자격 참고

대회 일정

접수 2026. 9. 7(월) 09:00 ~ 9. 22(화) 17:00

예선 2026. 10. 16(금) ~ 10. 17(토)
※ 온라인 진행

본선 2026. 11. 13 (금) | 리허설 및 연습 경기
2026. 11. 14 (토) | 본 경기 및 시상식
※ 오프라인 진행 / **일산 킨텍스 제2전시장**

참가 방법

ICPC Seoul Regional 에서 참가 접수

<https://seoul2026.icpcckorea.org>

시상 내역

대상	대통령상	1팀	상장 및 상금 500만원
금상	국무총리상	1팀	상장 및 상금 300만원
은상	과학기술정보통신부장관상	3팀	상장 및 상금 100만원
동상	한국정보사회진흥원장상	3팀	상장 및 상금 70만원
동상	한국정보과학회장상	1팀	상장 및 상금 70만원
장려상	한국정보사회진흥원장상	4팀	상장 및 상금 50만원
장려상	한국정보과학회장상	1팀	상장 및 상금 50만원

※ 시상내역은 변경될 수 있으며, 수상 자격은 국내 대학교 참가 팀에 한 함. 시상내역은 대회 동수에 따라 결정되며, 세계대회 참가 자격은 ICPC 지역대회 규정 선행.
※ 대회 결과에 따라 세계대회 참가자격 획득한 경우, 부상으로 상금 또는 세계대회 참가 경비 지원.

문의사항

☎ 070-5066-1963
✉ acmicpc@icpcckorea.org

홈페이지

QR코드 스캔 시
ICPC 홈페이지로
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과학기술정보통신부
Ministry of Science and ICT

NIA 한국지능정보사회진흥원
NATIONAL INFORMATION SOCIETY AGENCY



ASSIGNMENT #1 IS OUT

- Due : 9/17 (목) 23 : 59
- 주의 : 제출 기한이 지난 후에는 추가 제출 (딜레이)은 받지 않음





지난 시간 복습

RECURRENCE

1. Recurrence

2. Asymptotic analysis of recursive algorithms

- Iterative substitution
- Guess and verification
- Master theorem





목차

목차

1. Sorting – overview
2. Bubble sort
3. Selection sort
4. Insertion sort

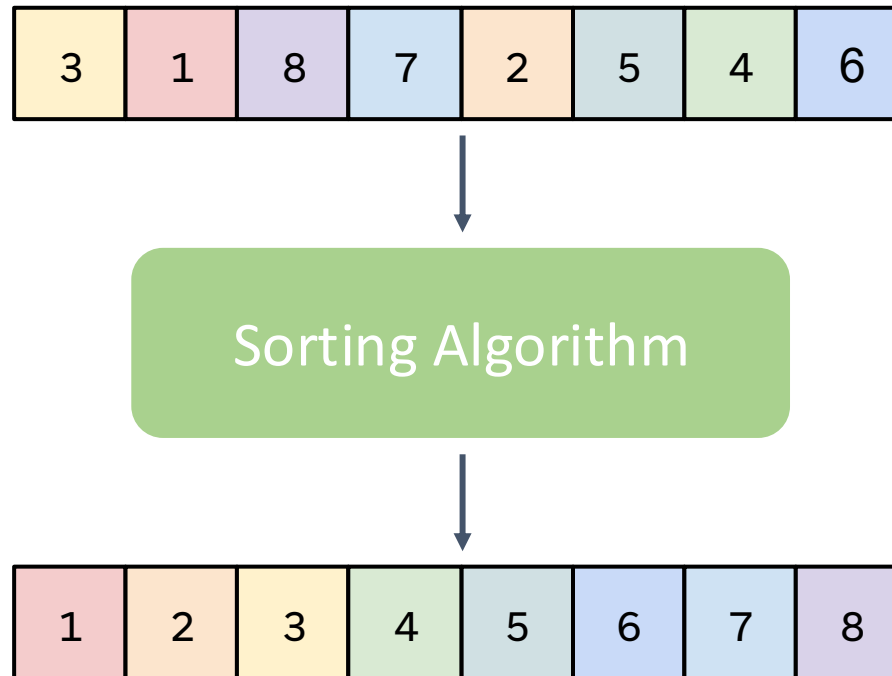


An abstract geometric design featuring two thin, dark lines that intersect on a light gray background. One line is oriented diagonally from the top-left towards the bottom-right, while the other is oriented more steeply from the top-left towards the bottom-right, crossing the first line.

SORTING - OVERVIEW

SORTING

- 정렬(Sorting): 배열 원소들을 번호 순, 알파벳 순 등 정해진 순서대로 재배치하는 문제
- 다양한 응용: 점수, 문서, 검색 결과, ...



SORTING ALGORITHMS

- **기본적인 정렬 알고리즘**

- 평균 시간복잡도가 $\Theta(n^2)$ 인 알고리즘들
- Selection Sort, Insertion Sort, Bubble Sort

- **고급 정렬 알고리즘**

- 평균 시간복잡도가 $\Theta(n \log n)$ 인 알고리즘들
- Merge Sort, Quicksort, Heapsort

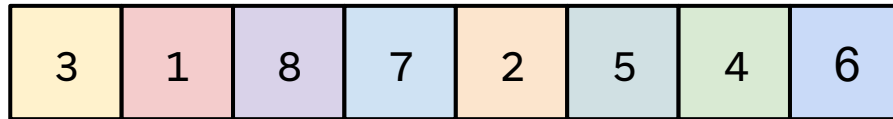
- **특수 정렬 알고리즘**

- 특수한 경우 시간복잡도가 $\Theta(n)$ 인 알고리즘들
- Radix Sort, Counting Sort

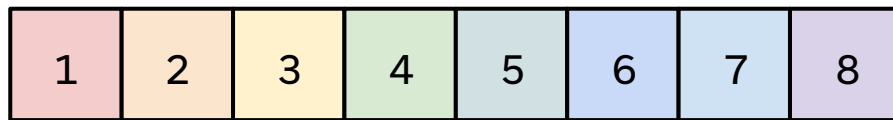


WHY SHOULD WE LEARN?

Q) 정렬 내장 함수가 있는데 굳이 배워야 할까?



정렬 내장 함수



```
nums = [3,1,8,7,2,5,4,6]
nums.sort()
print(nums)

>> [1,2,3,4,5,6,7,8]
```



WHY SHOULD WE LEARN?

Q) 정렬 내장 함수가 있는데 굳이 배워야 할까?

A) Yes !!



An abstract graphic design featuring two thin, dark gray lines that intersect on a light gray background. One line is oriented diagonally from the top-left towards the bottom-right, while the other is oriented from the top-right towards the bottom-left. The intersection point is located in the upper-left quadrant of the image.

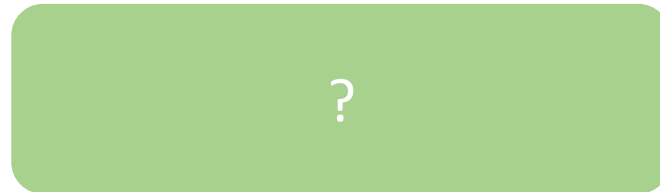
BUBBLE SORT

BUBBLE SORT

어떻게 정렬할 것인가?

cf) 오름차순 정렬 기준으로 설명

3	4	6	2	7	1	5	8
---	---	---	---	---	---	---	---



1	2	3	4	5	6	7	8
---	---	---	---	---	---	---	---



BUBBLE SORT

Intuition) 정렬이 완료된 시점을 기준으로

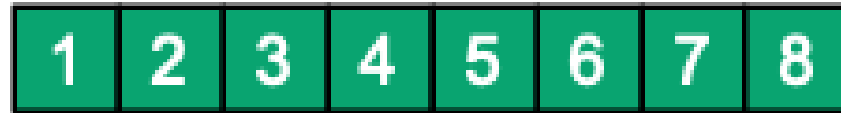
모든 인접한 배열 원소는 (왼쪽 원소 < 오른쪽 원소) 를 만족함



BUBBLE SORT

Intuition) 정렬이 완료된 시점을 기준으로

모든 인접한 배열 원소는 (왼쪽 원소 < 오른쪽 원소) 를 만족함

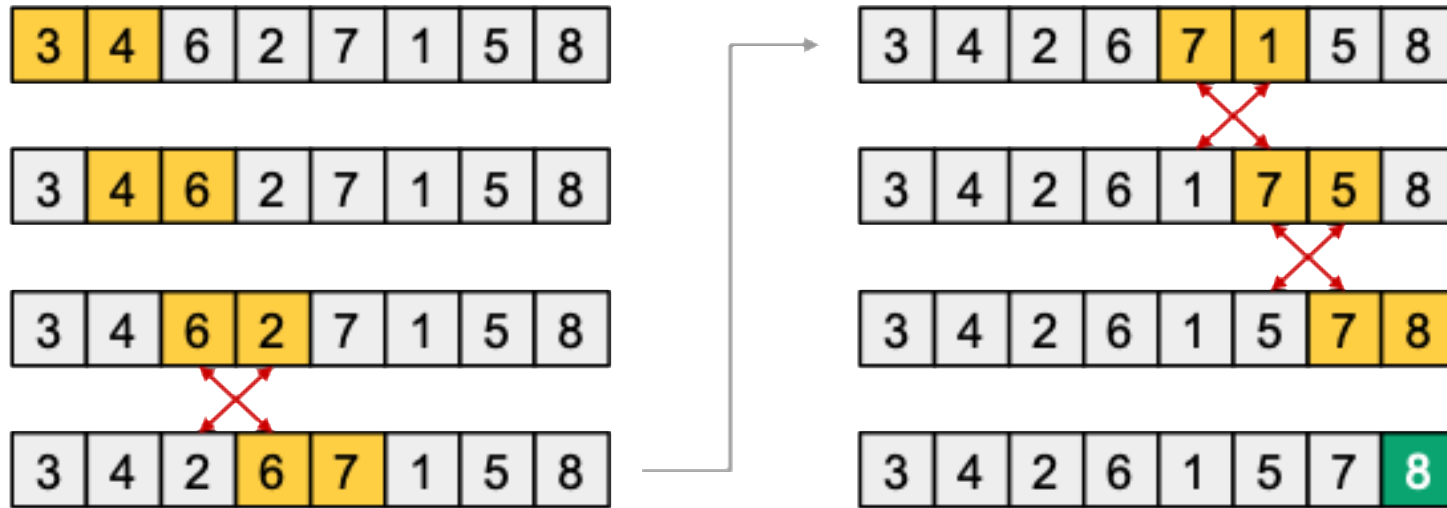


➔ 해당 조건이 위배되는 경우 위치를 서로 뒤바꾸면 다시 조건 만족



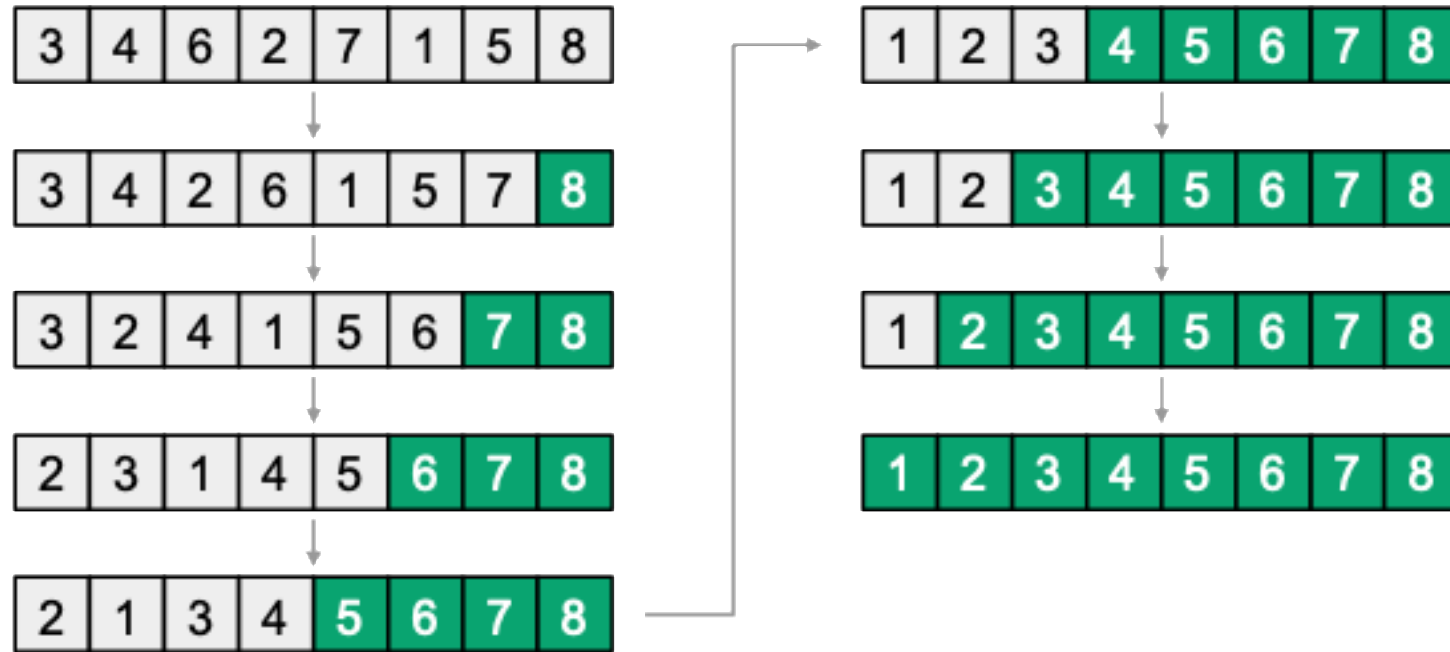
BUBBLE SORT

배열의 인접한 두 원소를 반복하여 비교 (Compare) 및 교환 (Swap)



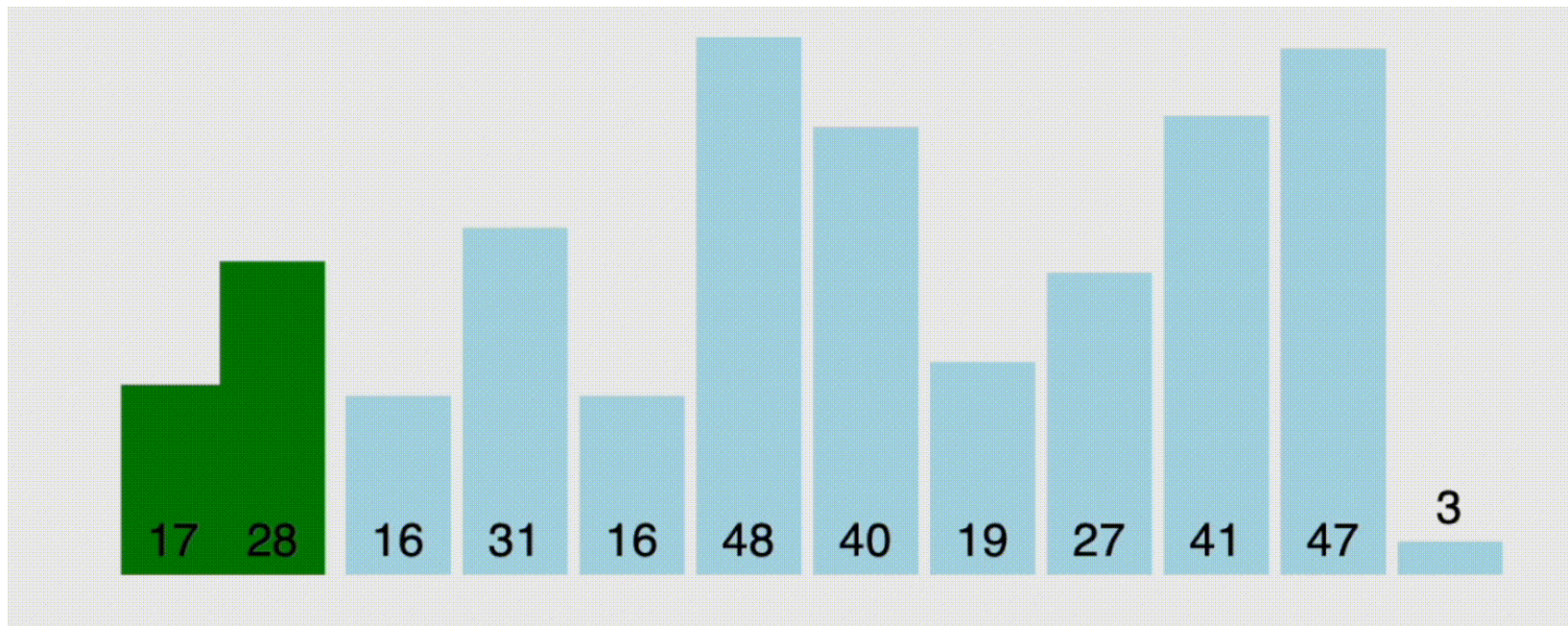
BUBBLE SORT

배열의 인접한 두 원소를 반복하여 비교 (Compare) 및 교환 (Swap)



BUBBLE SORT

Animation



BUBBLE SORT

Pseudo-code

```
def bubble_sort(arr):  
    n = len(arr)  
    for i in range(n-1,0,-1):  
        for j in range(i):  
            if arr[j] > arr[j+1]:  
                arr[j], arr[j+1] = arr[j+1], arr[j]
```

Cf)

3	4	6	2	7	1	5	8
---	---	---	---	---	---	---	---



BUBBLE SORT

Asymptotic analysis

	# swap	# comparisons	Complexity
Best Case	0	$\Theta(n^2)$	$\Theta(n^2)$
Worst Case	$\Theta(n^2)$	$\Theta(n^2)$	$\Theta(n^2)$
Average Case	$\Theta(n^2)$	$\Theta(n^2)$	$\Theta(n^2)$

Cf)

3	4	6	2	7	1	5	8
---	---	---	---	---	---	---	---



An abstract graphic design featuring two thin, dark gray lines that intersect on a light gray background. One line is oriented diagonally from the top-left towards the bottom-right, while the other is oriented from the top-right towards the bottom-left. The intersection point is located in the upper-left quadrant of the image.

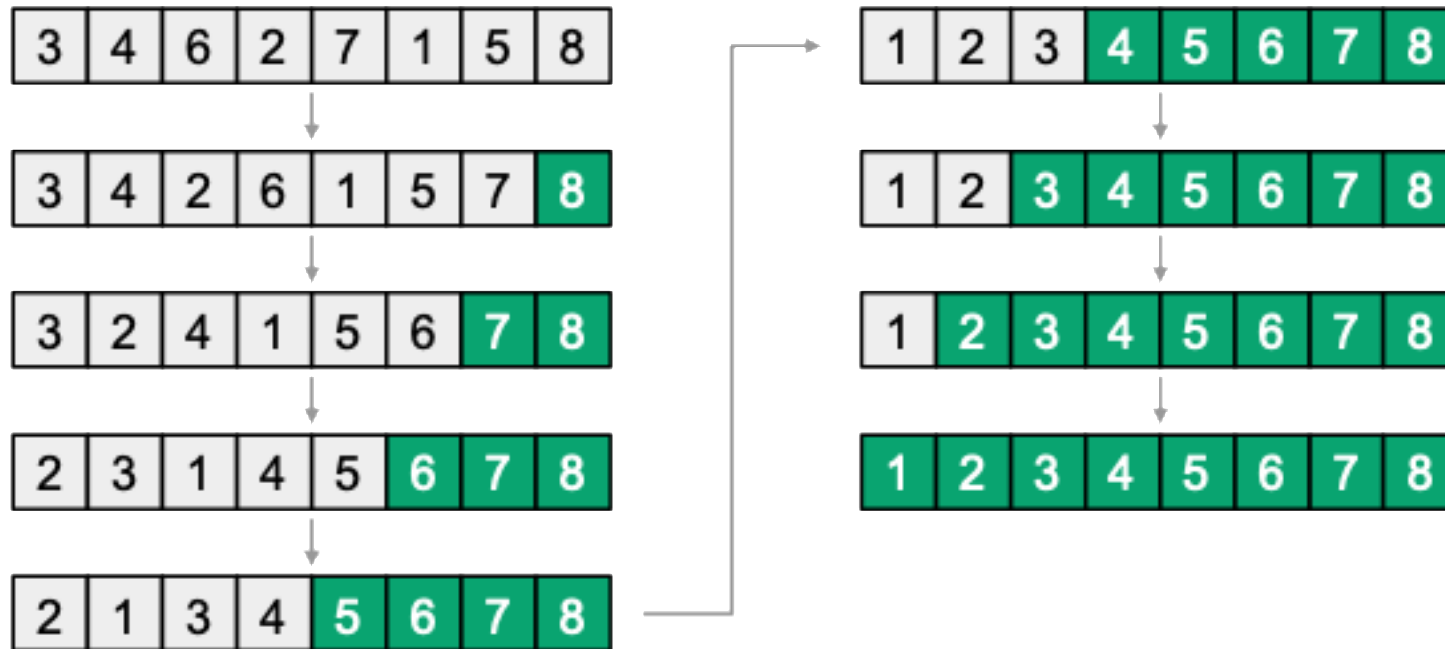
SELECTION SORT

SELECTION SORT

Intuition) 배열 원소 하나하나씩 올바른 위치를 찾아가게 하자

Cf) Divide and conquer

Ex) 버블정렬 - (선택정렬에서는 반대로 생각)

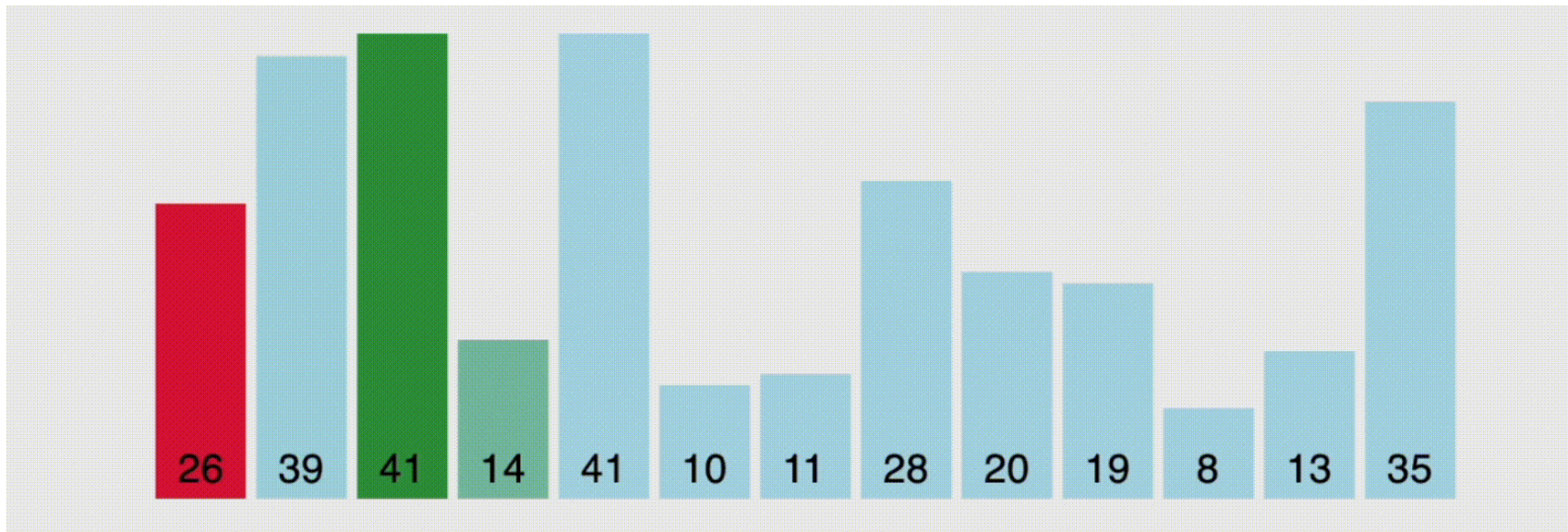


정렬되지 않은 배열 원소들 중에서 최솟값을 “찾아” 앞에 놓기 반복



SELECTION SORT

Animation



SELECTION SORT

Pseudo-code

```
def selection_sort(arr):  
    n = len(arr)  
    for i in range(n):  
        min_idx = i;  
        for j in range(i+1,n):  
            if arr[j] < arr[min_idx]:  
                min_idx = j  
        arr[i], arr[min_idx] = arr[min_idx], arr[i]
```

Cf)

3	4	6	2	7	1	5	8
---	---	---	---	---	---	---	---



SELECTION SORT

Asymptotic analysis

	# swap	# comparisons	Complexity
Best Case	0	$\Theta(n^2)$	$\Theta(n^2)$
Worst Case	$\Theta(n)$	$\Theta(n^2)$	$\Theta(n^2)$
Average Case	$\Theta(n)$	$\Theta(n^2)$	$\Theta(n^2)$

Cf)

3	4	6	2	7	1	5	8
---	---	---	---	---	---	---	---



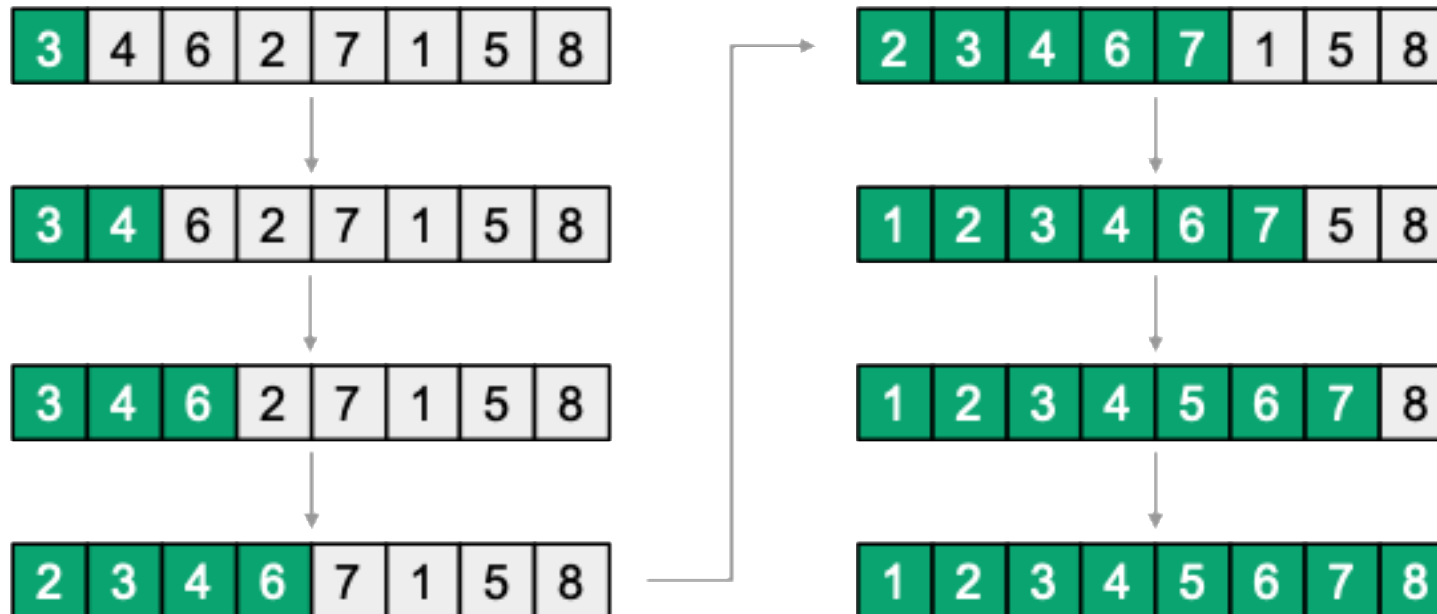
An abstract graphic design featuring two thin, dark gray lines that intersect on a light gray background. One line is oriented diagonally from the top-left towards the bottom-right, while the other is oriented from the top-right towards the bottom-left. The intersection point is located in the upper-left quadrant of the image.

INSERTION SORT

INSERTION SORT

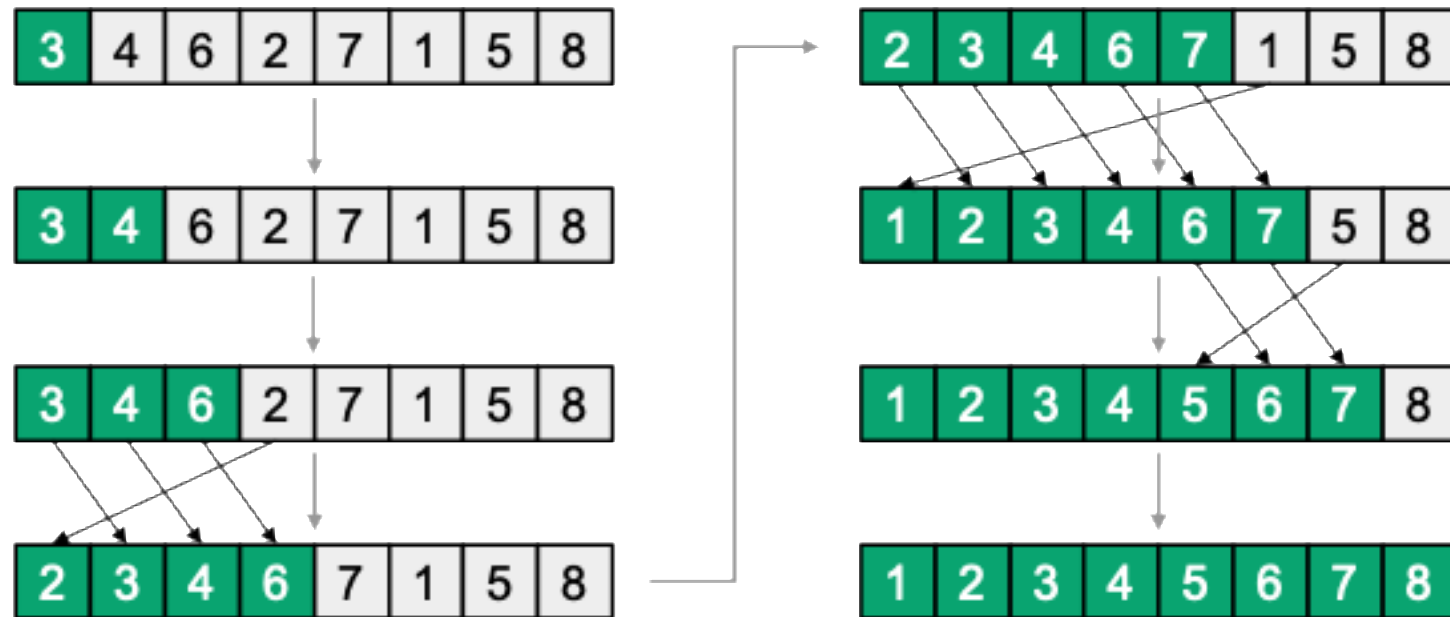
Intuition) 다른 관점에서의 Divide and conquer

배열 크기를 하나씩 늘려가면서 작은 정렬 문제를 여러 번 풀기



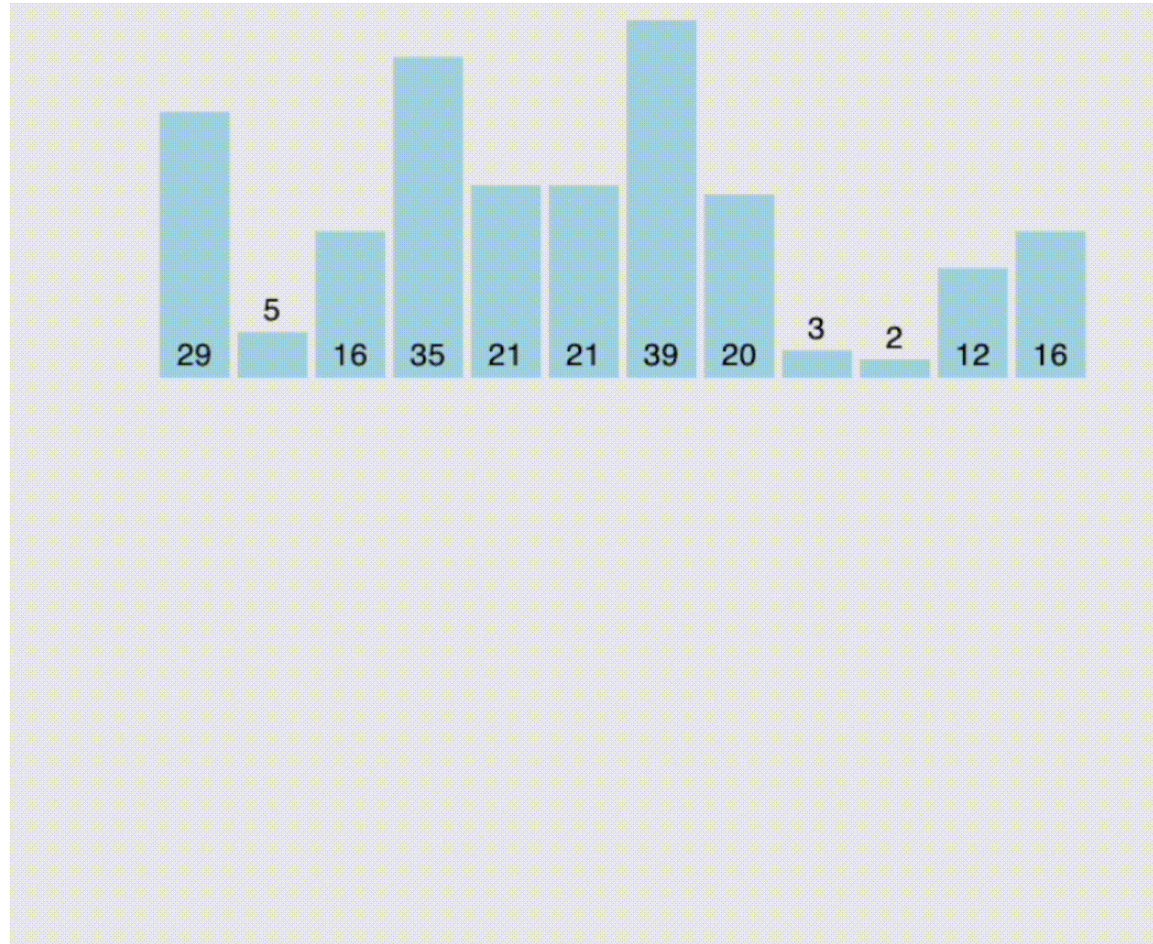
INSERTION SORT

정렬되지 않은 배열의 다음 원소를 이미 정렬된 값들 사이에 “끼워넣기” 반복



INSERTION SORT

Animation



INSERTION SORT

Pseudo-code

```
def insertion_sort(arr):  
    n = len(arr)  
    for i in range(1,n):  
        for j in range(i,0,-1):  
            if arr[j-1] <= arr[j]: break  
            arr[j-1], arr[j] = arr[j], arr[j-1]
```

Cf)

3	4	6	2	7	1	5	8
---	---	---	---	---	---	---	---



INSERTION SORT

Asymptotic analysis

	# swap	# comparisons	Complexity
Best Case	0	$\Theta(n)$	$\Theta(n)$
Worst Case	$\Theta(n^2)$	$\Theta(n^2)$	$\Theta(n^2)$
Average Case	$\Theta(n^2)$	$\Theta(n^2)$	$\Theta(n^2)$

Cf)

3	4	6	2	7	1	5	8
---	---	---	---	---	---	---	---

➔ 배열이 거의 정렬된 상태일 때 사용하면 매우 효율적



An abstract graphic on a light gray background. Two thin, dark gray lines intersect. One line is nearly vertical, starting from the top center and extending towards the bottom right. The other line is nearly horizontal, starting from the top left and extending towards the middle right. The intersection point is located in the upper-left quadrant of the image. To the right of the intersection, the word "SUMMARY" is written in a bold, black, sans-serif font.

SUMMARY

SUMMARY

1. **Sorting – overview**
2. **Bubble sort**
3. **Selection sort**
4. **Insertion sort**



NEXT LECTURE

Sorting (2) - Advanced sorting algorithms



Two thin, dark gray lines intersect diagonally on a light gray background. One line slopes downwards from left to right, and the other slopes upwards from left to right, crossing it at approximately one-third of the way across the frame.

QUESTIONS?